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6 authors, including:


Saif Agha
IRTA Institute of Agrifood Research and Technology


Andrea Doeschl-Wilson
The Roslin Institute


Curtis E Lind
The Commonwealth Scientific and Industrial Research Organisation


John Benzie
University College Cork

Genetic variation in microenvironmental plasticity of body weight in Tilapia (*Oreochromis niloticus*)

S. Agha[1,2], A. Doeschl-Wilson[1], W. Mekkawy[2,3], C. E. Lind[3], J. Benzie[3] and N. Ibanez-Escriche[1]

[1]The Roslin Institute, University of Edinburgh, Edinburgh, UK, [2]Animal Production Department, Faculty of Agriculture, Ain Shams University, Cairo, Egypt, [3]WorldFish, Penang, Malaysia, [4]Rajiv Gandhi Center for Aquaculture, Tamil Nadu, India. Safsoof23@hotmail.com

In animal breeding, genetic differences in environmental variation can be detected as genetic heterogeneity of residual variance which is also a measure of the micro-environmental plasticity (sensitivity). The objectives of this study were to (1) estimate the genetic variation in micro-environmental plasticity for body weight at harvest (BW) in Genetically Improved Farmed Tilapia strain (GIFT), and (2) investigate the effect of selective breeding for BW across multiple generations on its environmental variance. The data consist of 74,253 individuals representing 1131 full-sibs families selected for BW over 13 generations in Malaysia. Three different animal models were implemented using Bayesian inference. Model HOM assumes the homogeneity of environmental variance, while in model HET1 is assumed heterogeneous environmental variance and in model HET2 is assumed that part of the environmental variance is under genetic control. Deviance Information Criterion values favored HET2 model. The analysis revealed the presence of the additive genetic variance in both the mean (0.45) and the variance of BW (0.34). The posterior mean of the genetic additive correlation between the additive genes affecting the mean BW and its variance (95% highest posterior interval) was -0.53 (-0.47, -0.59). Negative correlation implies that genotypes with high BW tend to be less sensitive to local environmental perturbations (more homogeneous) i.e. robust. The genetic trends showed that the BW was improved by selective breeding, although not a clear correlated genetic change occurred in the environmental variance during the selection period. Including the micro-environmental sensitivity information in aquaculture breeding programs may improve both the performance and robustness of the farmed fish.

Keywords: Genetic heterogeneity of residual variance, environmental sensitivity, GIFT, aquaculture.